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1 Introduction

I arrived at Santiago the 9 of june. The goal of the trip was to continue learning research in game theory, with Roberto a mathematician specialized in optimization. We where supposed to work on congestion games but finally we settled on MDP's and reinforcement learning.

Reinforcement Learning (RL) is a branch of artificial intelligence that studies how an agent can learn to make optimal decisions in a given environment through trial and error. Unlike supervised learning, where correct input-output pairs are provided, RL focuses on learning from delayed feedback: the agent acts, observes the consequences of its actions, and gradually adjusts its strategy to maximize cumulative reward.

Formally, RL problems are often modeled as Markov Decision Processes (MDPs). An MDP is defined by a set of states, a set of possible actions, transition probabilities between states, and a reward function. The agent's goal is to determine an optimal policy, that is, a rule assigning the best action to each state in order to maximize either the total or average reward.

Reinforcement Learning has a wide range of applications across various fields. In robotics, RL allows machines to learn complex tasks, such as walking or grasping objects, by interacting with their environment. In finance, it is used for portfolio management and trading strategies, where the system adapts to changing market conditions. RL also plays a key role in recommendation systems, healthcare for personalized treatment strategies, and in traffic and transportation, where it can optimize routing and congestion management.

Having done some internship in game theory and working with Roberto, who has published extensively in the field, we didn't really see it as AI research but more as a decision-making problem.

The first part of the internship was spent reading literature to deepen my understanding of the subject. One interesting aspect of this phase was the diversity of the mathematics involved. For example, it was my first time working with max-plus algebra.

The second part focused on research. Basically, Roberto had published an algorithm that solved MDPs, and the goal of the research was to prove its optimality. We tried applying arguments from learning theory, and the work bordered on theoretical computer science. We exclusively attempted to find proofs, no simulations or data were involved.

2 MDP's

2.1 Definitions & Properties

Markov Decision Process

A **Markov Decision Process (MDP)** is a quadruple (S, A, P, R) , where:

- S is a finite set of states,
- A is a finite set of actions,
- $P : S \times A \times S \rightarrow [0, 1]$,
- $R : S \times A \rightarrow \mathbb{R}$ (can be $R : S \times S \times A \rightarrow \mathbb{R}$)

The game plays as follows:

- start at state s_0 ,
- at each time $t \in \mathbb{N}$ do the following:
 - choose an action a_t
 - the next state is given following the probability law $P(\cdot | s_t, a_t)$
 - get the reward $R(s_t, a_t)$ (or $R(s_{t+1}, s_t, a_t)$)

Policy and generated Markov chains

A **mixed policy** π is a mapping:

$$\pi : S \rightarrow \Delta(A), \quad \text{where } \Delta(A) \text{ denotes the set of probability distributions over } A.$$

A **policy** π is a mapping $\pi : S \rightarrow A$.

Given a policy π and a starting point s_0 we get the following Markov chains for the canonical filtration of s_t :

- the states s_t^{π, s_0} defined by $s_0^{\pi, s_0} = s_0$ and $s_t^{\pi, s_0} \sim P(\cdot | s_{t-1}^{\pi, s_0}, \pi(s_{t-1}^{\pi, s_0}))$

We then get the following sequences:

- the actions $a_t^{\pi, s_0} = \pi(s_t^{\pi, s_0})$
- the reward $r_t^{\pi, s_0} = R(s_t^{\pi, s_0}, a_t^{\pi, s_0})$

Type of MDP's

The long-term behavior of a MDP depends on the structure of the recurrent classes of its policies:

- A MDP is **recurrent** if all policies $\pi : \mathcal{S} \rightarrow \mathcal{A}$ induce a recurrent Markov chain.
- A MDP is **unichain** if all policies $\pi : \mathcal{S} \rightarrow \mathcal{A}$ induce a unichain Markov chain, *i.e.*, a Markov chain with a unique recurrent class, plus a (possibly empty) set of transient states.
- A MDP is **weakly communicating** if there exists a subset of states $\mathcal{R} \subset \mathcal{S}$ such that
 - (1) for all $s, s' \in \mathcal{R}$, there exists a path of positive probability of going from s to s' ¹, and
 - (2) the states $\mathcal{S} \setminus \mathcal{R}$ are transient for all policies.²

Gain and gain optimality

We define the "gain at time T " random variable given by $g_T^\pi(s) = \sum_{t=0}^T r_t^{\pi,s}$.

We then define the gain of π as the long-time average return:

$$g^\pi(s) = \lim_{T \rightarrow +\infty} \frac{1}{T+1} \mathbb{E}[g_T^\pi(s)] = \lim_{T \rightarrow +\infty} \frac{1}{T+1} \sum_{t=0}^T \mathbb{E}[r_t^{\pi,s}] \in \mathbb{R}.$$

We say that π^* is **gain optimal** for a MDP, if for every π and every s , $g^{\pi^*}(s) \geq g^\pi(s)$. **A gain optimal policy exists for every MDP.** We note $g^* = g^{\pi^*}$, which is the gain vector of π^* indexed by s .

$$\text{We have } g^*(s) = \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} P(s'|s, a) g^*(s').$$

For a recurrent or unichain MDP, the gain of a policy does not depend on the initial state. For a weakly communicating MDP, g^* does not depend on the initial state.

Bias and bias optimality

For π gain optimal, we define the "bias at time t " random variable given by $h_t^\pi(s) = \sum_{i=0}^t (r_i^{\pi,s} - g^*(s))$.

¹There is a policy π such as the Markov chain $s_t^{\pi,s}$ leads to state s' with strictly positive probability

²Text from Gast et al. (2023)

We then define the bias of π as the Cesaro limit of the bias at time t :

$$h^\pi(s) = \lim_{T \rightarrow +\infty} \frac{1}{T+1} \mathbb{E} \left[\sum_{t=0}^T h_t^\pi(s) \right].$$

We say that π is **bias optimal** for a MDP, if it is gain optimal and for every π' and every s , $h^\pi(s) \geq h^{\pi'}(s)$.

We note h^π , the bias vector of π indexed by s .

$$h^\pi(s) = R(s, \pi(s)) + \sum_{s' \in S} P(s'|s, \pi(s)) h^\pi(s')$$

Bellman's equations and Bellman's optimality

The Bellman's equations are the following:

$$g(s) = \max_{a \in A} \sum_{s' \in S} P(s'|s, a) g(s'), \text{ and } h(s) + g(s) = \max_{a \in A} \{R(s, a) + \sum_{s' \in S} P(s'|s, a) h(s')\},$$

with $g, h : S \rightarrow \mathbb{R}$

The only g for which they are solutions is g^* . We note $\mathcal{H} \neq \emptyset$, the set of h such as $h(s) = \max_{a \in A} \{R(s, a) + \sum_{s' \in S} P(s'|s, a) h(s')\} - g^*(s)$

We say that a policy π is **Bellman optimal**, if there exists a $h \in \mathcal{H}$ such as

$$\pi(s) \in \operatorname{argmax}_{a \in A} \{R(s, a) + \sum_{s' \in S} P(s'|s, a) h(s')\}$$

We define the operator $\mathcal{T}$ as $\mathcal{T}(h) = \max_{a \in A} \{R(s, a) + \sum_{s' \in S} P(s'|s, a) h(s')\}$

This comes from dynamical programming.

Optimality recapitulation

For a MDP:

A policy π is **Gain optimal**, if for every π and every s , $g^{\pi^*}(s) \geq g^\pi(s)$.

A policy π is **Bellman optimal**, if there exists a $h \in \mathcal{H}$ such as

$$\pi(s) \in \operatorname{argmax}_{a \in A} \{R(s, a) + \sum_{s' \in S} P(s'|s, a) h(s')\}$$

A policy π is **bias optimal**, if it is gain optimal and for every π' and every s , $h^\pi(s) \geq h^{\pi'}(s)$.

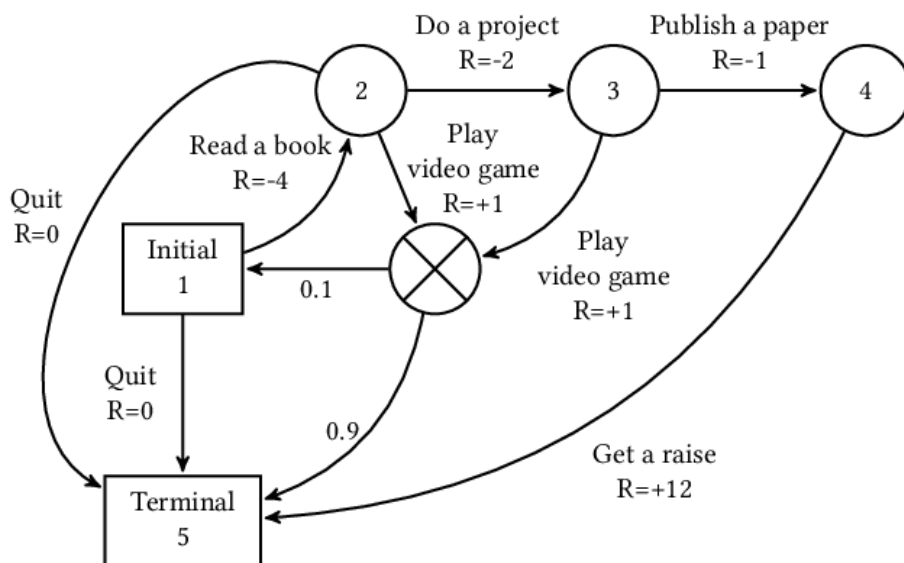
In general

$$\text{Gain-optimal} \supset \text{Bellman-optimal} \supset \text{bias-optimal}$$

but for unchain MDP's we can find example such as Gain-optimal $\supsetneq$ Bellman-optimal and Bellman-optimal $\supsetneq$ Bias-optimal

Example

Here is a fun example of an MDP i found [here](#) (I guess it is a minimization case).



Additional Comments

- We only look at infinite horizon.
- There is also infinite state MDP's.
- Here we don't talk at all about the discounted optimality or Blackwell optimality.
- The main reference on MDP is Puterman (2014).

2.2 Proofs

A lot of theses proofs are in Puterman 2014, but they are not all self contained. The goal here is to follow all the citations to try to make self contained proofs.

The proofs will be made for finite weakly communicating MDP's.

Lemma: we can find W and Q such as $P^\pi = W^{-1} \begin{bmatrix} Q & 0 \\ 0 & I \end{bmatrix} W$. If m is the number of recurrent node we have the following properties: I is a $m \times m$ matrix and Q a $|S| - m \times |S| - m$ one. 1 is not a eigenvalue of Q . $I - Q$ is invertible. $\sigma(Q) < 1$.

Proof: We can reorganize S to have P^π in the flowing canonical form

$$\begin{bmatrix} Q_0 & Q_1 & Q_2 & Q_3 & \cdots \\ & C_1 & 0 & 0 & \\ 0 & 0 & C_2 & 0 & \\ & 0 & 0 & C_3 & \\ & & & & \ddots \end{bmatrix}$$

with C_k the transition probability for the recurrent communicating classes and the Q for the transient states.

The proof uses mainly linear algebra and can be found in LAMOND et al. 1989. □

Fact: The gain converges

Proof: We will look at the Markov Chain generated by π with transition matrix P^π . We define the S vector $r^\pi = (R(s, \pi(s)))_s$. We then have the following

$g^\pi(s) = \lim_{T \rightarrow +\infty} \frac{1}{T+1} \sum_{t=0}^T e'_s(P^\pi)^t r^\pi$. It is thus sufficient to show convergence of $\frac{1}{T+1} \sum_{t=0}^T (P^\pi)^t$ in any operator norm.

From the lemma, we can find W and Q such as $P^\pi = W^{-1} \begin{bmatrix} Q & 0 \\ 0 & I \end{bmatrix} W$. With $I - Q$ is invertible and $\sigma(Q) < 1$.

following Puterman,

$$\frac{1}{T+1} \sum_{t=0}^T (P^\pi)^t = W^{-1} \begin{bmatrix} \frac{1}{T+1} \sum_{t=0}^T Q^t & 0 \\ 0 & I \end{bmatrix} W$$

$[\sum_{t=0}^T Q^t](I - Q) = I - Q^{T+1}$, so $\frac{1}{T+1} \sum_{t=0}^T Q^t = \frac{1}{T+1}(I - Q^{T+1})(I - Q)^{-1}$. But $\sigma(Q) = \lim_{T \rightarrow +\infty} \|Q^T\|_{\infty}^{\frac{1}{T}}$, we can deduce that Q^T is bounded. By the definition of limit, $\exists T, \epsilon > 0, \forall t \geq T, 0 \leq \|Q^t\|_{\infty}^{\frac{1}{t}} \leq 1 - \epsilon$, so $\|Q^T\|_{\infty} \rightarrow 0$, thus bounded. It then follows that $\frac{1}{T+1} \sum_{t=0}^T Q^t \xrightarrow{\|\cdot\|_{\infty}} 0$.

So $\frac{1}{T+1} \sum_{t=0}^T (P^{\pi})^t \rightarrow W^{-1} \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} W = P^{\pi\infty}$ and $g^{\pi}(s) = e'_s P^{\pi\infty} r^{\pi}$ exists and is finite. $\square$

Fact: The bias converges.

Proof: Similarly as the gain case,

$$\begin{aligned} H_T^{\pi} &:= \frac{1}{T+1} \sum_{t=0}^T \sum_{i=0}^t ((P^{\pi})^i - P^{\pi\infty}) = \frac{1}{T+1} \sum_{t=0}^T \sum_{i=0}^t W^{-1} \begin{bmatrix} Q^i & 0 \\ 0 & 0 \end{bmatrix} W \\ &= W^{-1} \begin{bmatrix} \frac{1}{T+1} \sum_{t=0}^T \sum_{i=0}^t Q^i & 0 \\ 0 & 0 \end{bmatrix} W \end{aligned}$$

But by telescoping, $\frac{1}{T+1} \sum_{t=0}^T \sum_{i=0}^t Q^i (I - Q)^2 = \frac{1}{T+1} \sum_{t=0}^T (I - Q^{t+1})(I - Q)$
 $= \frac{1}{T+1} [(T+1)(I - Q) + \sum_{t=0}^T (Q^{t+1} - Q^{t+2})] = I - Q + \frac{1}{T+1} (Q - Q^{T+2})$

But Q^{T+2} is bounded for every norm, so $\frac{1}{T+1} \sum_{t=0}^T \sum_{i=0}^t Q^i \rightarrow (I - Q)^{-1}$. We deduce from this that $H_T^{\pi} \rightarrow W^{-1} \begin{bmatrix} (I - Q)^{-1} & 0 \\ 0 & 0 \end{bmatrix} W = H^{\pi}$ and that $h^{\pi}(s) = e'_s H^{\pi} r^{\pi}$ exists and is finite. $\square$

Note that H^{π} is the Drazin inverse of $I - P^{\pi}$.

Fact: $g^{\pi}(s) = \sum_{C_i \in CR(P^{\pi})} P(\exists T, s_T^{\pi, s_0} \in C_i) g^{\pi}(s_i)$, with $s_i \in C_i$, where $CR(P^{\pi})$ is the set of communicating recurrent classes.

Proof: We define the following stopping time $\tau^{\pi, s} = \min_t \{t, s_t^{\pi, s} \in \bigcup C_i\}$. If s is in a communicating recurrent class of P^{π} , $\tau^{\pi, s} = 0$. If not, s is a transitive state of P^{π} . We thus have $P(\tau^{\pi, s} < \infty) = 1$.

$$g_T^{\pi}(s) = \mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s}] = \mathbb{E}[\mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau]] = \sum_{k \geq 0} P(\tau = k) \mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau = k]$$

We will look at $\lim_{T \rightarrow +\infty} \mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau = k]$ and conclude with the Dominated Convergence Theorem.

$$\mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau = k] = \frac{1}{T+1} \sum_{t=0}^{k-1} \mathbb{E}[r_t^{\pi, s} \mid \tau = k] + (1 - \frac{k}{T+1}) \frac{1}{T+1-k} \sum_{t=0}^{T-k} \mathbb{E}[r_{k+t}^{\pi, s} \mid \tau = k]$$

But $|\frac{1}{T+1} \sum_{t=0}^{k-1} \mathbb{E}[r_t^{\pi, s} \mid \tau = k]| \leq \frac{k}{T+1} R$ with $R = \max_{s \in S} |r^\pi(s)|$, so $\frac{1}{T+1} \sum_{t=0}^{k-1} \mathbb{E}[r_t^{\pi, s} \mid \tau = k] \rightarrow 0$.

In addition, by Markov property, $\mathbb{E}[r_{k+t}^{\pi, s} \mid \tau = k] = \mathbb{E}[r_t^{\pi, s_{\tau=k}}]$. So

$$\begin{aligned} \mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau = k] &= o_T(1) + (1 - \frac{k}{T+1}) \sum_{s_0 \in \bigcup C_i} P(s_k^{\pi, s} = s_0 \mid \tau = k) [\frac{1}{T+1-k} \sum_{t=0}^{T-k} \mathbb{E}[r_t^{\pi, s_0}]] \\ &\rightarrow \sum_{s_0 \in \bigcup C_i} P(s_k^{\pi, s} = s_0 \mid \tau = k) g(s_0) \end{aligned}$$

And $|\mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau = k]| \leq R$ measurable. We can then apply the Dominated Convergence Theorem. So

$$\begin{aligned} \lim_{T \rightarrow +\infty} \mathbb{E}[\mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau]] &= \mathbb{E}[\lim_{T \rightarrow +\infty} \mathbb{E}[\frac{1}{T+1} \sum_{t=0}^T r_t^{\pi, s} \mid \tau]] = \mathbb{E}[\sum_{s_0 \in \bigcup C_i} P(s_k^{\pi, s} = s_0 \mid \tau = k) g(s_0)] \\ &= \sum_{s_0 \in \bigcup C_i} \mathbb{E}[P(s_k^{\pi, s} = s_0 \mid \tau = k)] g(s_0) = \sum_{s_0 \in \bigcup C_i} P(s_\tau^{\pi, s} = s_0) g(s_0) \end{aligned}$$

To finish we will show that g is constant on each C_i . We look at $P_{C_i}^\pi$ the Markov chain restricted to C_i . It is still a Markov chains because it is a recurrent communicating class so the process gets out of the chains with proba 0. Because this chain is recurrent, there is a unique invariant probability measure w for $P_{C_i}^\pi$ and $\lim_{T \rightarrow +\infty} \frac{1}{T+1} \sum_{t=0}^T e'_s(P^\pi)^n \rightarrow w$. So $\forall s \in C_i, g(s) = w r^\pi$.

We then get that

$$g(s) = \sum_{s_0 \in \bigcup C_i} P(s_\tau^{\pi, s} = s_0) g(s_0) = \sum_{C_i} \sum_{s_0 \in C_i} P(s_\tau^{\pi, s} = s_0) g(s_i) = \sum_{C_i} P(s_\tau^{\pi, s} \in C_i) g(s_i)$$

□

This proof uses probability to show another way of looking at MDP's, but there also exists a proof with linear algebra like in LAMOND et al. 1989.

Fact: g^* exists and is constant on S for weakly communicating MDP's.

Proof: There exists π and s such as $\forall \pi', s', g^\pi(s) \geq g^{\pi'}(s')$ because the sets are finite. We can pick s recurrent because if s is transient for π , there exist s_2 recurrent under π such as $g^\pi(s) \leq g^\pi(s_2)$ which is a consequence of

$$g^\pi(s) = \sum_{C_i \in CR(P^\pi)} P(\exists T, s_T^{\pi, s_0} \in C_i) g^\pi(s_i), \text{ with } s_i \in C_i$$

We will define C as the recurrent class of s . If $S \setminus (T \cup C) = \emptyset$ then this is a gain optimal strategies. If $S \setminus (T \cup C) \neq \emptyset$ we will construct recursively a strategy π^* with only one final class C .

We will set π^* equal to π on S . Because, $S \setminus (T \cup C) \neq \emptyset$, there exists $s_C \in S \setminus (T \cup C)$ and $a_1 \in A$ such as $\exists s_1 \in C, P(s_C | s_1, a_1) > 0$ because the MDP is weakly communicating. We set $\pi^*(s_1) = a_1$. We define $C_1 = C \cup \{s_1\}$. $\forall k$, if $S \setminus (T \cup C_k) \neq \emptyset$, we choose $s_{k+1} \in S \setminus (T \cup C_k)$ such as $\exists s_{C_k} \in C_k, a_{k+1} \in A, P(s_{C_k} | s_{k+1}, a_{k+1}) > 0$. We set $\pi^*(s_{k+1}) = a_{k+1}$ and $C_{k+1} = C_k \cup \{s_{k+1}\}$. We do this recursively until $S \setminus (T \cup C_K) = \emptyset$. The action taken on T doesn't matter.

C is still recurrent on π^* . For every point in $s_k \in S \setminus (T \cup C)$ there is a positive prob that the process goes into C_{k-1} . at each point this the case so there is positive prob path from any point to C . We deduce that C is the only recurrent communicating class of P^π . So $g(s') = P(s_\tau^{\pi, s} \in C)g(s) = g(s)$. $\square$

Fact: - If $g\mathbf{1}$ and h satisfies WC Bellman equitation then $g = g^*$

- Gain-optimal $\supset$ Bellman-optimal

Proof: $h(s) + g = \max_{a \in A} \{R(s, a) + \sum_{s' \in S} P(s' | s, a)h(s')\}$

$\Leftrightarrow g\mathbf{1} = \max_{\pi} \{r_\pi + (P^\pi - I)h\}$. It follows that for every policy π , we have $g\mathbf{1} = P^\pi g\mathbf{1} \geq P^\pi r_\pi + P^\pi (P^\pi - I)h$. We the same argument $g\mathbf{1} \geq (P^\pi)^T r_\pi + (P^\pi)^T (P^\pi - I)h$. By summing we have the following result the inequalities, we have $(T+1)g\mathbf{1} \geq \sum_{t=0}^T (P^\pi)^t r_\pi + \sum_{t=0}^T (P^\pi)^t (P^\pi - I)h$. following arguments for the convergence of the gain, $\sum_{t=0}^T (P^\pi)^t (P^\pi - I) = (P^\pi)^{T+1} - I$, so $(T+1)g\mathbf{1} \geq \sum_{t=0}^T (P^\pi)^t r_\pi + ((P^\pi)^{T+1} - I)h$. We also know that $\frac{1}{T+1}((P^\pi)^{T+1} - I) \rightarrow 0$ in every norm. So $g\mathbf{1} \geq \lim_{T \rightarrow +\infty} \frac{1}{T+1} \sum_{t=0}^T (P^\pi)^t r_\pi = g^\pi$. So $g \leq g^*$.

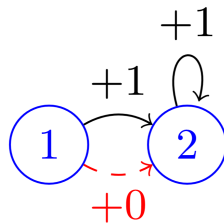
We note π^B , the policy that reaches the max of $r_\pi + (P^\pi - I)h$. π^B exists because there is a finite set of polices. We then have $g\mathbf{1} = r_{\pi^B} + (P^{\pi^B} - I)h$. With the same method we can show that $g^{\pi^B} = g$. $\square$

Fact: $\mathcal{T}(h) - g^*$ has a fixed point

Proof: See Puterman (2014) p448. □

Fact: There exists MDP's with Gain-optimal $\supsetneq$ Bellman-optimal.

Proof: This counter-example is for Gast et al. 2023. It is a WC deterministic MDP.



For any $h \in \mathcal{H}$, $h(1) = \max_{a \in \{a, \bar{a}\}} \{R(1, a) + h(2)\} - 1$ car $g^* = 1$. So $h(1) = \max\{h(2), h(2) - 1\}$.
 Donc $\bar{a} \notin \operatorname{argmax}_{a \in \{a, \bar{a}\}} \{R(1, a) + h(2)\}, \forall h$. So the policy π where you play red in 1 is not Bellman-optimal. But it is still gain optimal. □

In the next chapter we will see a characterization of Gain optimal policies in relation with $\mathcal{H}$.

2.3 Structure of $\mathcal{H}$ and its relation with h^π

The goal of this chapter is to give some context on $\mathcal{H}$ and how it relates to the policies. The explanation that is sometimes given for Bellman's equation is that for π gain optimal, we have $h^\pi(s) = R(s, \pi(s)) + \sum_{s' \in S} P(s'|s, \pi(s))h^\pi(s')$. And because it is optimal it would be more or less $\max_{a \in A} R(s, a) + \sum_{s' \in S} P(s'|s, a)h^\pi(s')$. But we will see that in general $h^\pi \notin \mathcal{H}$ and that the link between gain optimal policies and Bellman is a bit more complicated.

There will be less proofs in this section because a lot of the material is self contained and because it is less essential to our work.

Form and dimension of $\mathcal{H}$

This is from Akian, Bapat, et al. 2007.

For a Deterministic MDP we can take $S = A$. If there is no action from s to s' we define $R(s, s') = -\infty$ if there is an action we define $R(s, s') = \max_{a \in A, P(s'|s, a)=1} R(s, a)$. This MDP has the same optimal strategies and takes the same paths.

We say that a cycle $(s_0, s_1, \dots, s_k, s_0)$ is critical if $\frac{1}{k} [\sum_{0 \leq t < k} R(s_t, s_{t+1}) + R(s_k, s_0)] = g^*$. The union of the graph of the critical cycles is the critical graph, its nodes are the critical nodes N^c and its communicating classes are the critical class C^c of the MDP. We then construct the distance matrix A^+ where $A^+_{s, s'} = \max_{(s_0=s, s_1, \dots, s_k=s')_{0 \leq s < k}} [R(s, s_1) + \dots + R(s_k, s_0) - g^*]$. It represents the best payoff relative to g^* going from s to s' . It is finite because there are finitely many states and the value on a cycle c is lower than g^* .

We then have the following result:

$$\mathcal{H} = \{h, h(s) = \max_{s' \in N^c} [\lambda_{s'} + A^+_{s, s'}], \lambda_{s'} \in R\}$$

We also have for s and s' in N^c that $A^+_{s, s} - A^+_{s', s'} = \lambda \mathbf{1} \Leftrightarrow s$ and s' are in the same critical class. **We can then deduce that $\mathcal{H}$ is a convex set of dimension the number of critical classes.**

This problem is the eighteen problem in max plus algebra. To see the proofs and the details we invite you to look at Akian, Bapat, et al. 2007.

A generalization to random MDP's of this result exists in Akian and Gaubert 2003.

h^π and $\mathcal{H}$

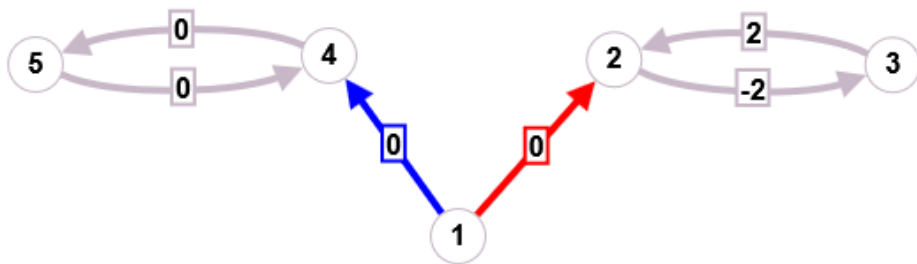
Fact: There are MDP's and gain optimal strategies such as $h^\pi \notin \mathcal{H}$.

Proof: In the case of example LIENNN, $h^\pi(1) = -1 \neq \arg\max_{a \in \{a, a\}} \{R(1, a) + h(2)\} - 1 = 0$.
 In this case $g^* = 1$, π is gain optimal, his bias vector is $(-1, 0)$. $\square$

But a lot of algorithm focus on finding Bellman optimal strategies. But in this case also things don't work.

Fact: There are MDP's and Bellman optimal strategies such as $h^\pi \notin \mathcal{H}$.

Proof: Consider the flowing deterministic MDP:



Consider the function h with $h(1) = h(2) = 4$, $h(3) = 6$ and $h(4) = h(5) = 0$. $h \in \mathcal{H}$ and $\pi(1) \in \arg\max_{a \in \{a, a\}} \{h(s(a))\}$, with π the policy where you take the red path in 1, so π is Bellman optimal.

But $h^\pi(1) = -1 \neq \max\{h(2), h(4)\}$. So $h^\pi \notin \mathcal{H}$. $\square$

Gain optimality and $\mathcal{H}$

Finally from Gast et al. 2023, we get the link between gain optimal policy and $\mathcal{H}$.

Fact: Let π be a policy and let $\mathcal{R}^\pi$ be the set of recurrent states of the Markov chain induced by π .

Then, π is gain-optimal if and only if: there exists $h \in \mathcal{H}$ such as

- For all $s \in \mathcal{S}$, $\pi(s)$ attains the max in $\max_{a \in A} \sum_{s' \in \mathcal{S}} P(s'|s, a)g(s')$.
- $\pi(s)$ attains the max in $\max_{a \in A} \{R(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a)h(s')\}$ for all $s \in \mathcal{R}^\pi$.

3 Our work

3.1 Goal and definitions

3.1.1 Definitions

A big part of the research addresses algorithm that find approximations of gain optimal policy.

In this case we consider algorithms that are given the states, the actions and the reward function, but not the transition probability. Instead, To estimate them, you are allowed to sample from a generative model: you can query the model for a number of realizations of $P(s' | s, a)$. The goal is to design an algorithm that minimizes the number of samples while still producing an approximate solution with high probability.

To formalize what is efficiency, we have the following definitions:

Definition: Given an MDP(**with rewards between 0 and 1**) whose state transitions can only be accessed by sampling, an (ϵ, δ) -PAC-MDP algorithm is an algorithm $\mathcal{A}$ such that with probability at least $1 - \delta$ computes a policy π such that $g^* - g^\pi(s) < \epsilon$ for all states $s \in S$.

Note that some algorithm can work only on a class of MDP's.

Definition: We say that an (ϵ, δ) -PAC-MDP algorithm $\mathcal{A}$ has a sample complexity $O(f)$ if for each (ϵ, δ) , with probability at least $1 - \delta$ the number of samples used by the algorithm is bounded by some constant times f , where f a function that depends both on (ϵ, δ) as well as the MDP characteristics R, S, A, H .

Here R denotes the rewards, S is the number of states, A the number of actions, and H an upper bound for some solution h of Bellman's equation

Definition: $\|x\|_{sp} = \max_s x_s - \min_s x_s$

3.1.2 Our goal

Roberto, my internship supervisor, co-authored a paper Lee et al. 2025 with Jongmin Lee and Mario Bravo describing an algorithm that finds an ϵ -gain optimal policy with high probability. The goal of our work was to prove that the complexity of their algorithm could not be improved asymptotically. In their paper they describe a (ϵ, δ) -PAC algorithm. That has a proven average sample size of $O(|S||A|\epsilon^{-2}\|h\|_{sp}^2 \ln(\frac{1}{\delta}))$, where h is a Bellman solution for the MDP. The question is: is there an algorithm with a better sample complexity in this class of algorithms? We mainly supposed this has optimal and tried to prove the optimality.

The algorithm is based on fixed point iteration on Bellman equation for WC MDP's. It is based on the literature for fixed point iteration on non-expensive function ie function such as $\|f(x) - f(y)\| \leq \|x - y\|$. Because, the Bellman operator is not necessary contracting but is non-expensive for the ∞ norm. The details of the algorithm are not necessary to prove optimality but there are details in the Lee et al. 2025.

3.1.3 Formalization of an algorithm

In this part we will adrese what is exactly an algorithme in this case. The following definitions will be overly precises but i prefer having a global idea and relaxing the definition in fear of missing some key assumptions.

In this context an algorithm is a (possibly random) turing machine with acces to the generative model.

Algorithms with prior knowledge are given an upper bound of the same order of the span norm of one of the bellman solution of the MDP in input. Algorithms with out prior knowledge are not.

Definition: We say that an (ϵ, δ) -PAC-MDP algorithm $\mathcal{A}$ has a space complexity $O(f)$ if, for each (ϵ, δ) , with probability at least $1 - \delta$, the amount of memory used by the algorithm is bounded by f up to constant factors. Here, f is a function that depends on (ϵ, δ) as well as on the MDP characteristics R, S, A, H .

We say that an algorithm is **model based** if it has a space complexity of $O(|S|^2|A|)$. Because you can model the hole MDP. We say that an algorithm is **model free** if it has a space complexity of $O(|S||A|)$.

3.1.4 Simplifications

There are several ways to simplify what we call an algorithm. This is an important part because it allows to do prof easily and focus on a certain properties of the algorithms.

Simplifications 1

For any $\mathcal{A}$, we get 3 functions nxm, f_m, f_π (or probability distribution). nxm gives the next move (a node and an action to sample or the "stop" instruction), f_m changes the memory after each sample in this way: $M = f_m(M, s)$. And finally decides a policy starting with M . We get the following algorithm:

While $nxm(M) \neq stop$:
 $s = Sample(nxm(M))$
 $M = f_m(M, s)$

return $f_\pi(M)$

Every algorithm has the 3 functions, but 3 functions doesn't necessarily give you an algorithm because not every function can be simulated by a Turing Machine. But it makes some arguments easier. You can use probability distribution instead of functions if you want to use a random algorithm.

Simplification are important, many are possible. It may be worth it to work on this. I know that Roberto had one that i wanted to include, but i forgot the details.

3.1.5 Previous work

Now we will, see the previous work that have been done. Were is a recap of the important articles from Lee et al. 2025.

Table 1: Comparison of sample complexity results for average-reward MDPs

Previous works	Sample complexity	Method type	MDP class	Prior knowl
Zhang et al. 2023	$\tilde{O}(S A h^* _{\text{sp}}^2/\varepsilon^2)$	model-free	W	1
Zurek et al. 2024	$\tilde{O}(S A h^* _{\text{sp}}^2/\varepsilon^2)$	model-based	W	1
Tuynman et al. 2024	$\tilde{O}(S A D/\varepsilon^2)$	model-based	C	$\times$
Lee et al. 2025	$\tilde{O}(S A h^* _{\text{sp}}^2/\varepsilon^2)$	model-free	W	$\times$

Definition: We say that an (ϵ, δ) -PAC-MDP algorithm $\mathcal{A}$ has a sample complexity $\tilde{O}(f)$ if, for each (ϵ, δ) , with probability at least $1 - \delta$, the number of samples used by the algorithm is bounded by f up to logarithmic factors. Here, f is a function that depends on (ϵ, δ) as well as on the MDP characteristics R, S, A, H .

If the algorithm of Lee et al. 2025 is optimal in his class, what makes the class have $\tilde{O}(|S||A|||h^*||_{\text{sp}}^2/\varepsilon^2)$ at best? Is it the fact that it works for WC, the fact that it is model free, that it works with out prior knowledge or a combination of the three?. The answer to this question will give a direction for a prof.

3.2 Working with arguments from multi-armed Bandit

3.2.1 Wang et al. 2022

The first idea we explored to produce lower bounds comes from Wang et al. 2022. It comes from learning theory and multi armed bandit. This allows to prove that any algorithm does at least $\tilde{O}(|S||A|||h^*||_{\text{sp}}/\varepsilon^2)$ samples on average

The idea is to construct very similar MDP's but with different ϵ -optimal policies. Because they are very similar you need to sample a lot to "learn" in which MDP you are. So if you give an answer with too few samples your risk giving the same answer for another MDP with high probability.

Formally, the theorem goes as follow:

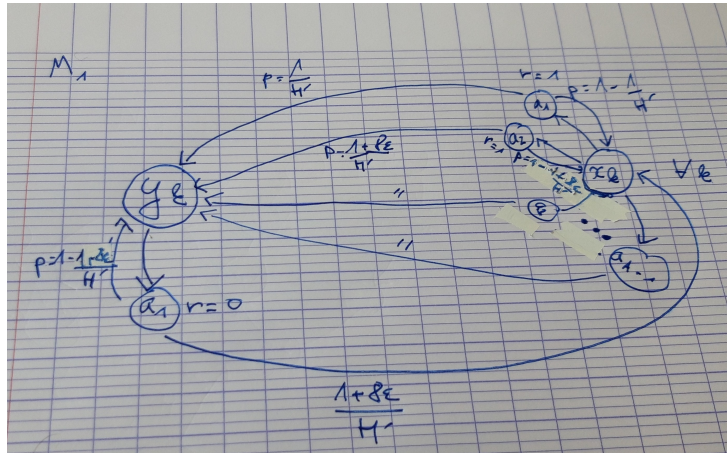
Theorem. For every $\epsilon < \frac{1}{16}, \delta < \frac{1}{4}, H > 4, A > 3, S > 2$, there exists a communicating MDP, M_1 (with S states A action, and a bellman solution with span H) for which every $\mathcal{A}(\epsilon, \delta)$ PAC-MDP, has an average sample size greater than $C|S||A|\epsilon^{-2}||h||_{\text{sp}}\ln(\frac{1}{\delta})$.

This is enough to show that every $\mathcal{A}(\epsilon, \delta)$ PAC-MDP that work for Communicating MDP's has a worst case sample complexity of at least $\tilde{O}(|S||A|||h^*||_{\text{sp}}/\varepsilon^2)$.

Let's set $(\epsilon, \delta), S, A, H. H' = H(2 + 8\epsilon)$

Here is the MDP of the theorem, M_1 :

First we will take a look at the core component, M_1 will be copies of the following MDP linked together.



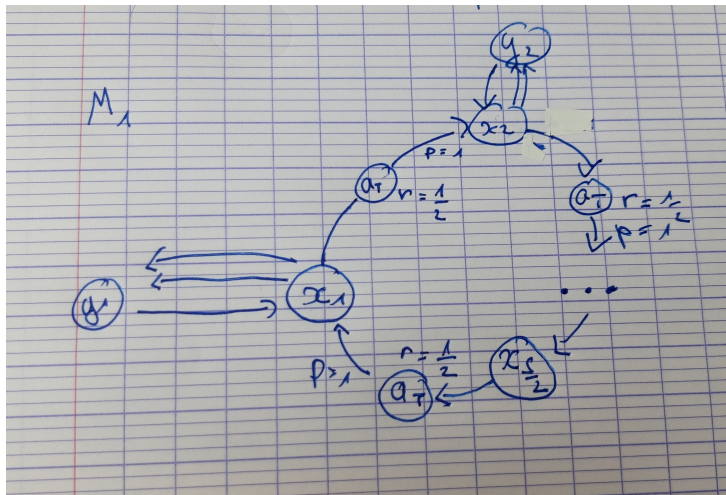
In the core component there is only to states x_k and y_k . In x_k , a_1 rewards 1 and $P(y_k|x_k, a_1) = \frac{1}{H'}$, $P(x_k|x_k, a_1) = 1 - \frac{1}{H'}$. All other actions in x_k ($a_i, 1 < i < A$), they reward 1 and $P(y_k|x_k, a_i) = \frac{1+8\epsilon}{H'}$, $P(x_k|x_k, a_i) = 1 - \frac{1+8\epsilon}{H'}$. In y_k all action give the same the reward 0, and $P(x_k|y_k, a) = \frac{1+8\epsilon}{H'}$, $P(y_k|y_k, a) = 1 - \frac{1+8\epsilon}{H'}$.

Basically the two policy possible on this core component, is to play a_1 or a_2 in x_k because all the actions in y_k are equivalent and all other actions in x_k are equivalent to a_2 . If we play a_1 , we have the following Markov chain:

$$\begin{pmatrix} 1 - \frac{1}{H'} & \frac{1}{H'} \\ 1 - \frac{1+8\epsilon}{H'} & \frac{1+8\epsilon}{H'} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1+8\epsilon}{2+8\epsilon} & \frac{1}{2+8\epsilon} \\ \frac{1+8\epsilon}{2+8\epsilon} & \frac{1}{2+8\epsilon} \end{pmatrix}$$

It converges because it is an aperiodic and recurring Markov chain. By definition of the gain, the policy where a_1 is played has gain $\frac{1+8\epsilon}{2+8\epsilon}$. Using the same method playing a_2 has gain $\frac{1}{2} \cdot \frac{1+8\epsilon}{2+8\epsilon} - \frac{1}{2} = \frac{8\epsilon}{2+8\epsilon} > \epsilon$ because $\epsilon < \frac{1}{16}$. So the only ϵ -solution is a_1 .

Now M_1 , is core components linked together in a cycle, like this:



To make it work we add an action a_t to go to the next x and rewards $\frac{1}{2}$. If you play $\pi(x_i) = a_t$, you get gain $\frac{1}{2}$, this policy isn't a ϵ -solution. On a x_k if you don't play a_t you have to play a_1 or else it isn't optimal as we saw in the core component. Basically all the ϵ -solution are the following: $\{\pi \text{ such as } \forall k, \pi(x_k) \in \{a_t, a_1\} \text{ and } \exists k, \pi(x_k) = a_1\}$.

You can check that h with $\forall k, h(x_k) = 0, h(y_k) = -H$ is a bellman solution, with span H .

Now $M_{k,l}$:

Given a k and a l , $M_{k,l}$ is the same MDP than M_1 excepte for the action a_l of the node x_k . In this case it still gives reward 1 but $P(y_k|x_k, a_l) = \frac{1-8\epsilon}{H'}$, $P(x_k|x_k, a_l) = 1 - \frac{1-8\epsilon}{H'}$. It is a very small change but it completely changes the ϵ -optimal solutions. In this case the only one is π , $\pi(x_k) = a_l$ and $\forall i \neq k, \pi(x_k) = a_t$. Indeed, this strategy gives $\frac{1}{2} + 4\epsilon$. And $\frac{1}{2} + 4\epsilon - \frac{1+8\epsilon}{2+8\epsilon} = \epsilon(4\frac{1+8\epsilon}{2+8\epsilon}) > \epsilon$.

Probability distribution given an algorithm

Let's give ourselves an algorithm $\mathcal{A}$. We define as P_1 , the probability distribution over the sampling behavior of the algorithm if it runs on M_1 and $P_{k,l}$ if it runs on $M_{k,l}$. They

are probability distribution over Ω finite sequence of samples (an action and a state plus the results from the simulation).

$$\Omega = \bigcup_{n=0}^{\infty} \{(s, a, s')\}^n,$$

We then define the random variables $N_{k,l}$, which is the number of sample on the node x_k and the action a_l . And $S_{k,l}$, is the number of sample on the node x_k and the action a_l which gave back x_k . Naturally, we have that $N_{k,l} - S_{k,l}$ gave y_k in this case.

Given an algorithm $\mathcal{A}$ and an element $w \in \Omega$, the likelihood of w ($L_1(w)$ if $\mathcal{A}$ runs on M_1 or $L_{k,l}(w)$ if $\mathcal{A}$ runs on $M_{k,l}$) design the the proba of the sequence happening.

$$\begin{aligned} L_1(w) &:= P_1(W = w) \\ &= \prod_{i=1}^{|w|} P(\mathcal{A} \text{ samples in } (s_i, a_i) | (w_k)_{k=1}^n) P_1(s'_i | s_i, a_i) \times P(\mathcal{A} \text{ stops at } w) \end{aligned}$$

With $w_i = (s_i, a_i, s'_i)$

To make the proof we first prove the flowing lemma:

Lemma 3.1. *For all algorithm $\mathcal{A}$ and $\forall k, l$, ($E_1[N_{k,l}] \leq LB$ and $P_1(E_k) \leq \delta$) $\Rightarrow P_{k,l}(E_k) < 1 - \delta$. With $E_k = \{\mathcal{A} \text{ outputs } \pi \text{ such as } \pi(x_k) = a_1 \text{ or } a_t\}$ and $LB = \frac{1}{2048} H \epsilon^{-2} \ln(\frac{1}{4\delta})$.*

Proof.

$$P_{k,l}(E_k) = E_{k,l}[\mathbb{1}_{E_k}] = E_1\left[\frac{L_{k,l}(W)}{L_1(W)} \mathbb{1}_{E_k}\right]$$

This is true because every event is possible with both MDP and it is a countable probability space.

Now we can work with the assumptions on P_1 and try to show that $P_{k,l}(E_k) < 1 - \delta$. The idea will be to upper bound $\frac{L_{k,l}(W)}{L_1(W)}$ for all $W \in E_k$ under some likely hypothesis.

$$\begin{aligned} A_{k,l} &:= \{N_{k,l} \leq 4LB\}, \\ C_{k,l} &:= \left\{ \max_{1 \leq N_{k,l} \leq 4LB} |p_0 \cdot N_{k,l} - S_{k,l}(N_{k,l})| \leq \sqrt{16LB \cdot p_0 \cdot (1 - p_0) \cdot \ln \frac{1}{4\delta}} \right\}. \\ \mathcal{E}_{k,l} &= E_k \cap A_{k,l} \cap C_{k,l} \end{aligned}$$

Let's show that $P_1(\mathcal{E}_{k,l}) \geq \frac{1}{4}$:

We first show that

$$\mathbb{P}_1(A_{k,l}) \geq \frac{3}{4} \text{ and } \mathbb{P}_1(C_{k,l}) \geq \frac{3}{4}$$

By Markov's inequality,

$$\mathbb{P}_1(A_{k,l}) = 1 - \mathbb{P}_1(\overline{A_{k,l}}) \geq 1 - \frac{\mathbb{E}_1[N_{k,l}]}{4LB} \geq 1 - \frac{LB}{4LB} = \frac{3}{4}.$$

By definition of $S_{k,l}$ and the MDPs, $S_{k,l}$ is the sum of $N_{k,l}$ i.i.d. Bernoulli $\left(1 - \frac{1+8\epsilon}{D'}\right)$ random variables. Moreover, $\left((1 - \frac{1+8\epsilon}{D'}) \cdot N_{k,l} - S_{k,l}(N_{k,l})\right)$ is a martingale. Using Doob's inequality on Martingales, we get

$$\mathbb{P}_1(C_{k,l}) = 1 - \mathbb{P}_1(\overline{C_{k,l}}) \geq 1 - \frac{\mathbb{E}_1\left[\left((1 - \frac{1+8\epsilon}{D'}) \cdot 4LB - S_{k,l}(4LB)\right)^2\right]}{16LB \cdot \frac{1+8\epsilon}{D'} \cdot \left(1 - \frac{1+8\epsilon}{D'}\right) \cdot \ln \frac{1}{4\delta}} = 1 - \frac{1}{4 \ln \frac{1}{4\delta}} \geq \frac{3}{4},$$

given that $\delta \leq \frac{1}{16}$.

So

$$\begin{aligned} P_1(\mathcal{E}_{k,l}) &= P_1(E_k) + P_1(A_{k,l}) + P_1(C_{k,l}) - P_1(E_k \cup (A_{k,l} \cap C_{k,l})) - P_1(A_{k,l} \cup C_{k,l}) \\ &> \frac{3}{4} + \frac{3}{4} + (1 - \delta) - 1 - 1 \geq \frac{1}{4} \end{aligned}$$

We can now upper bound $P_{k,l}(E_k)$ in the following way:

$$\begin{aligned} P_{k,l}(E_k) &= E_1\left[\frac{L_{k,l}(W)}{L_1(W)} \mathbb{1}_{E_k}\right] \geq E_1\left[\frac{L_{k,l}(W)}{L_1(W)} \mathbb{1}_{\mathcal{E}_{k,l}}\right] \\ &\geq E_1\left[\mathbb{1}_{\mathcal{E}_{k,l}} \inf_{w \in \mathcal{E}_{k,l}} \frac{L_{k,l}(w)}{L_1(w)}\right] = P_1(\mathcal{E}_{k,l}) \inf_{w \in \mathcal{E}_{k,l}} \frac{L_{k,l}(w)}{L_1(w)} \geq \frac{1}{4} \inf_{w \in \mathcal{E}_{k,l}} \frac{L_{k,l}(w)}{L_1(w)} \end{aligned}$$

It is then sufficient to show that $\forall w \in \mathcal{E}_{k,l}, \frac{L_{k,l}(w)}{L_1(w)} > 4\delta$.

Let take $w \in \mathcal{E}_{k,l}$, we then define $N := N_{k,l}(w)$ and $S := S_{k,l}(w)$.

$$\frac{L_{k,l}(w)}{L_1(w)} = \frac{\left(1 - \frac{1-8\epsilon}{H'}\right)^S \left(\frac{1-8\epsilon}{H'}\right)^{N-S}}{\left(1 - \frac{1+8\epsilon}{H'}\right)^S \left(\frac{1+8\epsilon}{H'}\right)^{N-S}}$$

Because every other part of the likelihood (3.2.1) simplifies. Ended, it is the same for everything except the sampling in x_k and a_l .

Because $w \in \mathcal{E}_{k,l}$, $N \leq 4LB$ and S such as $|pN - S| \leq \sqrt{16LBp(1-p)}$ with $p := 1 - \frac{1+8\epsilon}{H'}$. We also set $\Delta := \frac{16\epsilon}{H'}$. Note that $\Delta \leq \frac{1-p}{2}$ because $\epsilon \leq \frac{1}{16}$. All this allows us to use Lemma 4.1 to show:

$$\frac{L_{k,l}(w)}{L_1(w)} = \frac{(p + \Delta)^S (1 - p - \Delta)^{N-S}}{(p)^S (1 - p)^{N-S}} > \exp\left\{-2(4LB) \frac{\Delta^2}{p^2(1-p)} + \Delta \sqrt{\frac{16LB}{p(1-p)}}\right\} > 4\delta$$

□

Now for the proof of the theorem:

Proof. Let take $\mathcal{A}$ (ϵ, δ) PAC-MDP. All the ϵ -optimal strategy on M_1 plays a_1 or a_t on every x . The only ϵ -optimal strategy on $M_{k,l}$ plays a_l on x_k and t on other x . So, because $\mathcal{A}$ is (ϵ, δ) PAC-MDP, $P_1(E_k) < \delta$ and $P_{k,l}(E_k) \geq 1 - \delta$. Because of the lemma, $E_1[N_{k,l}] \leq LB$ is false, otherwise the implication would imply that $P_{k,l}(E_k) < 1 - \delta$.

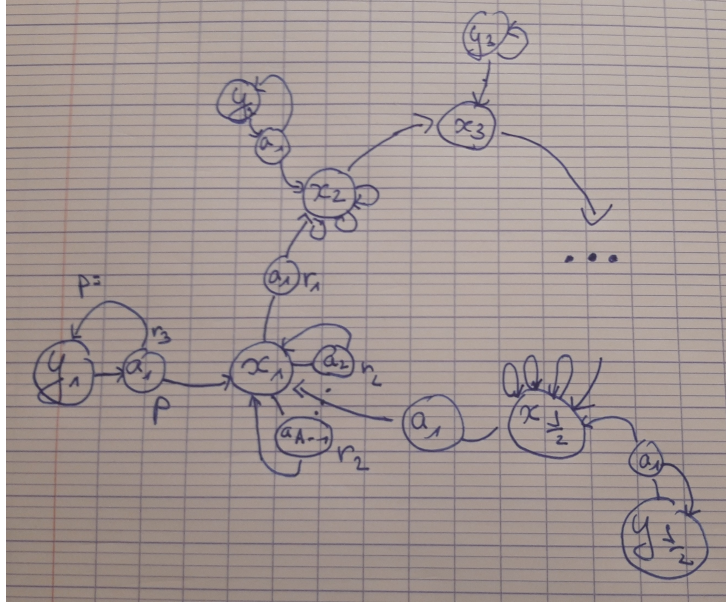
For every x and a , there are at least LB samples. So, overall, the algorithm samples at least $\frac{S}{2}(A-1)LB = \frac{1}{2048} \frac{S}{2}(A-1)\epsilon^{-2}H\ln(\frac{1}{\delta})$. $\square$

3.2.2 Adaptation to weakly communicating MDP's

The first research work and the most relevant work i did, has to use the previous idea but for Weakly Communicating MDP's. Maybe WC is just more complex than communicating and this is sufficient to show optimality of Lee et al. 2025. Instead of playing on the gain of the polycys, we will change transient state to communicating keeping the same idea.

We will build the proof, backwards by putting down constraint that the mdp need's to have to make the proof work.

First we will detail MDP, M_1 :



Like Wang et al. 2022, there is a core component that is linked in a cycle. Action a_1 on x_k , gives reward r_1 and brings to the next x . Action $a_i (i > 1)$ on x_k gives r_2 and keeps you on the same node. On y_k all action gives r_3 , and we have $P(x_k|y_y, a_i) = p$ and $P(y_k|y_y, a_i) = 1 - p$.

$M_{k,l}$ is the same but a_l on x_k still gives r_2 but $P(y_k|x_y, a_l) = q$ and $P(x_k|x_k, a_l) = 1 - q$.

The goal is to have the only ϵ -optimal policy in M_1 beeing playing a_1 in every x . And in $M_{k,l}$, the optimal move is to play a_l in x_k because you have a chance to get r_3 . Finally you want q small so it is hard to sample.

We need $r_1 - r_2 > \epsilon$ so that the only ϵ -optimal policy in M_1 is a_1 in every x_k . We also need $\frac{q}{p+q}r_3 + \frac{p}{p+q}r_2 - r_1 > \epsilon$ so that the only ϵ -optimal policy in $M_{k,l}$ is a_l in every x_k and a_1 in the other x . In this case in M_1 there is only one h Bellman up to a $\lambda 1$. We have $\|h\|_{sp} = \frac{r_3 - r_1}{p}$ in M_1 . Because we want the example to work for every span of h , we need $H = \frac{r_3 - r_1}{p}$, with H a parameters of the graphs.

Lemma 3.2. *For $\delta < \frac{1}{4}$, $q \leq \frac{1}{2}$, if $E_1[N_{k,l}] \leq LB$ and $P_1(E_k = \{\mathcal{A} \text{ gives } \pi, \pi(x_k) = a_1\}) \geq 1 - \delta$ then $P_{k,l}(E_k) > \frac{1}{2}e^{-8LBq}$.*

Proof. We take k and l . We build $B = \{N_{k,l} \leq 4LB\}$, we also define $E = E_k \cap B$. By Markov, $P_1(B) \geq 1 - \frac{1}{4}$, so $P_1(E) \geq \frac{3}{4} + 1 - \delta - 1 > \frac{1}{2}$.

$P_{k,l}(E_k) = E_{k,l}[\mathbb{1}_B] = E_1[\mathbb{1}_B \frac{l_{k,l}(W)}{l_1(W)}] \geq P(B) \min_{w \in B} \frac{l_{k,l}(w)}{l_1(w)}$. But $\frac{l_{k,l}(w)}{l_1(w)} = (1 - q)^N \geq (1 - q)^{4LB}$ in B . Because $q \leq \frac{1}{2}$, $(1 - q)^{4LB} \geq \exp\{-8LBq\}$. So $P_{k,l}(E_k) > \frac{1}{2}e^{-8LBq}$. $\square$

To make the argument work we need $\frac{1}{2}e^{-8LBq} \geq \delta \iff LB \leq \frac{\ln(\frac{1}{2\delta})}{8q}$. To get the biggest lower bound we take $LB = \frac{\ln(\frac{1}{2\delta})}{8q}$. But we also need

$$\frac{q}{p+q}r_3 + \frac{p}{p+q}r_2 - r_1 > \epsilon \Rightarrow \frac{qpH}{p+q} - \frac{p}{p+q}(r_1 - r_2) > \epsilon \Rightarrow qH - \epsilon(1 + \frac{q}{p}) > r_1 - r_2$$

. Because we also need $r_1 - r_2 > \epsilon$, we get that $qH - \epsilon(1 + \frac{q}{p}) > \epsilon \Rightarrow p(2\epsilon - qH) < -q\epsilon$. Because p , q and ϵ are positive, we need $p(2\epsilon - qH) < -q\epsilon$ implying $\frac{H}{2\epsilon} > \frac{1}{q}$. Thus with this example the best we can hope for is $LB \leq \frac{\ln(\frac{1}{2\delta})H}{16\epsilon}$. It is thus impossible to prove a H^2 lower bound for this MDP with this method. This is a negative result.

4 Conclusion

In the end, we didn't get any results. I had a lot of negatives. That being said I now have a better understanding of the field and of the difficulties of proving optimality.

A lot can be done to continue, the proof in Wang et al. 2022 was for algorithm that worked on communicating MDP, with prior knowledge and model based. I explored what happened if you extended the reasoning to WC, there is surely more to do, but I don't have ideas on this front. What I would do if I continued would be to explore arguments using the "without prior knowledge" component or the "model free" aspect. Memory complexity theory and Multi-armed bandit could be good reads.

But in retrospect, it was a lot of fun and I learned a lot.

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Annexes

Technical lemma for Wang2022

Lemma 4.1. Fix B , p and $|\Delta| \leq \frac{\min\{p, 1-p\}}{2}$, let $N \leq B$ and S such as $|pN - S| \leq \sqrt{cBp(1-p)}$ then $E := \frac{(p+\Delta)^S(1-p-\Delta)^{N-S}}{(p)^S(1-p)^{N-S}} \geq \exp\{-2(B\frac{\Delta^2}{p^2(1-p)} + \Delta\sqrt{\frac{cB}{p(1-p)}})\}$

Proof.

$$\begin{aligned} E &= (1 + \frac{\Delta}{p})^S (1 - \frac{\Delta}{1-p})^{N-S} \\ &= (1 + \frac{\Delta}{p})^S (1 - \frac{\Delta}{1-p})^{S(\frac{1}{p}-1)} (1 - \frac{\Delta}{1-p})^{N-\frac{S}{p}} \end{aligned}$$

Using $1 - u \geq e^{-u-u^2}$ and $e^{-u} \geq 1 - u$, for $u \in [0, \frac{1}{2}]$ with $\Delta \leq \frac{\min\{p, 1-p\}}{2}$ we get:

$$\begin{aligned} (1 - \frac{\Delta}{1-p})^{\frac{1}{p}-1} &= \exp\{\frac{1-p}{p} \ln(1 - \frac{\Delta}{1-p})\} \geq \exp\{\frac{1-p}{p} (-\frac{\Delta}{1-p} - (\frac{\Delta}{1-p})^2)\} \\ &= \exp\{-\frac{\Delta}{p} - \frac{\Delta^2}{(1-p)p}\} = \exp\{-\frac{\Delta}{p}\} \exp\{-\frac{\Delta^2}{(1-p)p}\} \\ &\geq (1 - \frac{\Delta}{p})(1 - \frac{\Delta^2}{(1-p)p}) \end{aligned}$$

So $E \geq (1 - (\frac{\Delta}{p})^2)^S (1 - \frac{\Delta^2}{(1-p)p})^S (1 - \frac{\Delta}{1-p})^{N-\frac{S}{p}} \geq (1 - (\frac{\Delta}{p})^2)^B (1 - \frac{\Delta^2}{(1-p)p})^B (1 - \frac{\Delta}{1-p})^{\sqrt{cB\frac{1-p}{p}}}$.

Using again the fact that $\ln(1 - u) \geq -2u$ for $0 \leq u \leq \frac{1}{2}$, we get that

$$\begin{aligned} E &\geq \exp\{-2B(\frac{\Delta}{p})^2 - 2B\frac{\Delta^2}{(1-p)p} - 2\sqrt{cB\frac{1-p}{p}} \frac{\Delta}{1-p}\} \\ &= \exp\{-2(B\frac{\Delta^2}{p^2(1-p)} + \Delta\sqrt{\frac{cB}{p(1-p)}})\} \\ &= \exp\{-2(\frac{\Delta}{p}\sqrt{\frac{B}{1-p}} + \frac{\sqrt{cp}}{2})^2 + \frac{cp}{2}\} \end{aligned}$$

□

Victorien Marette, Ensaë 2A
Stage d'application 2024-2025
Research internship – MDP
Pontificia Universidad Católica de Chile, Santiago de Chile
Internship supervisor: Roberto Cominetti
9 June – 15 September 2025

Objectif du stage

Le stage s'est déroulé à la *Pontificia Universidad Católica de Chile* à Santiago du 9 juin au 15 septembre 2025, sous la supervision de Roberto Cominetti. L'objectif principal était de continuer à apprendre la recherche en théorie des jeux et en optimisation, avec un focus sur les **processus de décision markoviens (MDP)** et l'apprentissage par renforcement.

Contexte général

L'**apprentissage par renforcement** est une branche de l'intelligence artificielle qui étudie comment un agent peut apprendre à prendre les meilleures décisions dans un environnement en expérimentant et en observant les conséquences de ses actions. - Contrairement à l'apprentissage supervisé, l'agent ne reçoit pas de réponse "correcte" après chaque action, mais doit apprendre à optimiser ses choix sur le long terme. - Les MDP sont des modèles mathématiques utilisés pour représenter ce type de problème, avec des **états**, **actions**, **probabilités de transition** et **récompenses**. L'objectif est de déterminer la meilleure stratégie pour maximiser la récompense cumulée.

Travail effectué pendant le stage

Le stage s'est organisé en deux grandes étapes :

1. **Étude de la littérature** : lecture approfondie sur les MDP et l'apprentissage par renforcement, et découverte de nouveaux outils mathématiques.
2. **Recherche** : objectif de démontrer l'optimalité d'un algorithme publié par Roberto Cominetti pour résoudre les MDP. Aucun calcul informatique n'était nécessaire : le travail portait sur des preuves théoriques.

Conclusion

Ce stage a été une expérience enrichissante pour approfondir les connaissances en prise de décision, optimisation et recherche théorique. Il a permis de combiner théorie mathématique et idées venant de l'intelligence artificielle, tout en renforçant la rigueur scientifique et les méthodes de démonstration.

Victorien Marette, Ensaе 2A
Applied Internship 2024-2025
Research internship – MDP
Pontificia Universidad Católica de Chile, Santiago, Chile
Internship supervisor: Roberto Cominetti
9 June – 15 September 2025

Internship Objective

The internship took place at the *Pontificia Universidad Católica de Chile* in Santiago from June 9 to September 15, 2025, under the supervision of Roberto Cominetti. The main objective was to continue learning research in game theory and optimization, with a focus on **Markov Decision Processes (MDPs)** and reinforcement learning.

General Context

Reinforcement learning is a branch of artificial intelligence that studies how an agent can learn to make the best decisions in an environment by experimenting and observing the consequences of its actions. - Unlike supervised learning, the agent does not receive a "correct" answer after each action, but must learn to optimize its choices over the long term. - MDPs are mathematical models used to represent this type of problem, with **states**, **actions**, **transition probabilities**, and **rewards**. The goal is to determine the optimal strategy to maximize cumulative reward.

Work Carried Out During the Internship

The internship was organized into two main stages:

1. **Literature Review**: in-depth reading on MDPs and reinforcement learning, and exploration of new mathematical tools.
2. **Research**: the goal was to demonstrate the optimality of an algorithm published by Roberto Cominetti for solving MDPs. No computational calculations were required; the work focused on theoretical proofs.

Conclusion

This internship was an enriching experience for deepening knowledge in decision-making, optimization, and theoretical research. It allowed combining mathematical theory with ideas from artificial intelligence, while strengthening scientific rigor and proof techniques.